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SIT225: Data Capture Technologies

Activity 2.1: Working with sensor - DHT22

DHT22 is a temperature and humidity sensor.

Hardware Required

Arduino Board

DHT22 sensor

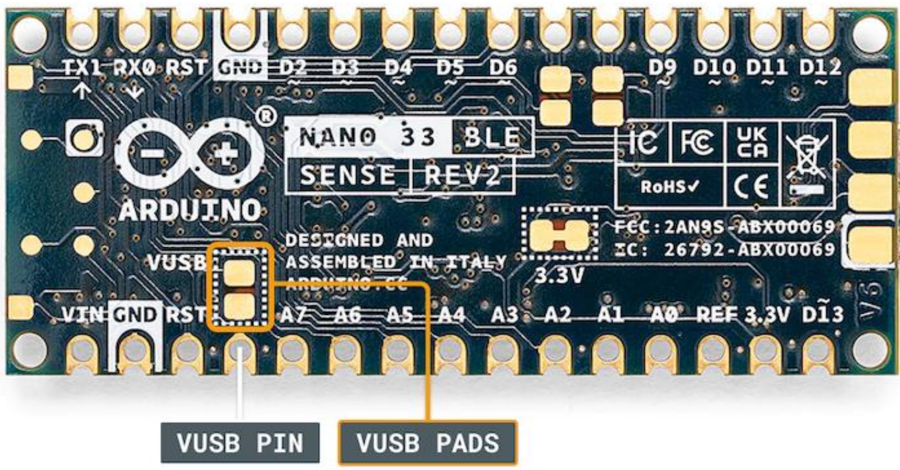
USB cable

Software Required

Arduino programming environment

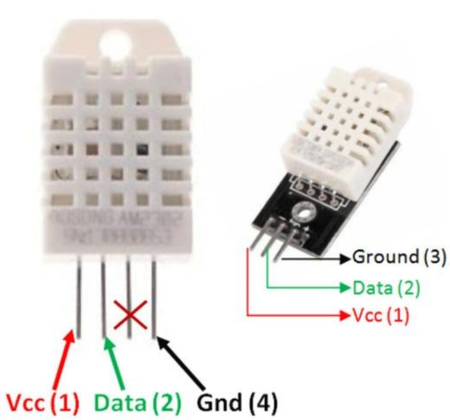
Known issue, action required

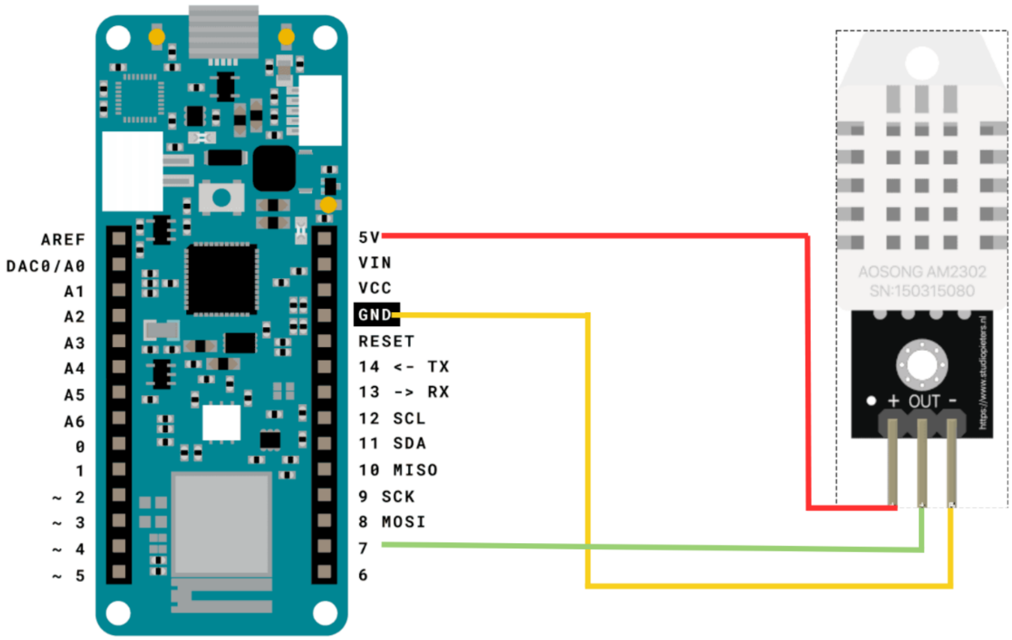
Arduino Nano 33 IoT board operates on 3.3 V, it needs to be arranged to make it 5 V. The Arduino board has a pin called VUSB or VBUS and there are 2 pads next to the pin. **These two pads must be shorted to enable the pin** (see detail here https://support.arduino.cc/hc/en-us/articles/360014779679-Enable-5-V-power-on-the-VUSB-or-VBUS-pin-on-Nano-boards).



To test, you can use a wire and connect these 2 pads manually by hand to see if data is coming through DHT22 sensor. For long data collection, you will need to solder the wire permanently to connect the pads. Seeking help from tutors there is an on-campus facility called Maker Space where you can do it.

Steps:

Step	Action
1	Connect your DHT22 Temperature and Humidity Sensor to the Arduino board. Note that the pin layout in the image below may look different than the board you may have. 

	 a. Pick a red male-female jumper wire and attach the female end to pin 1 (VCC pin) on the sensor. Plug the male end into the Arduino board's 5V power pin. b. Pick a blue male-female jumper wire and attach the female end to pin 2 (DATA pin) on the sensor. Plug the male end into the Arduino board's digital data pin 2. c. Pick a black male-female jumper wire and attach the female end to pin 4 (GND) on the sensor. Plug the male end into the Arduino board's GND pin. d. Sensor's pin 3 is not used.
2	Connect your Arduino board to your computer using the USB cable.
3	Write an Arduino sketch (or download it from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_2/sketch_dht22.ino) which looks like below. Compile the code in Arduino IDE, deploy to the board and observe output in the Arduino IDE serial monitor.

```
// Add "DHT sensor library"
#include <DHT.h>

#define DHTPIN 2  // digital pin number
#define DHTTYPE DHT22  // DHT type 11 or 22
DHT dht(DHTPIN, DHTTYPE);

// variables to store data.
float hum, temp;

void setup() {
  // Set baud rate for serial communication
  Serial.begin(9600);

  // initialise DHT libarary
  dht.begin();
}

void loop() {
  // read data
  hum = dht.readHumidity();
  temp = dht.readTemperature();

  // Print data to serial port - a long way
  //
  // Serial.print("Humid: ");
  // Serial.print(hum);
  // Serial.print(" %, Temp: ");
  // Serial.print(temp);
  // Serial.println(" Celsius");

  // Print data to serial port - a compact way
  Serial.println(String(hum) + "," + String(temp));

  // wait a while
  delay(15*1000);
}
```

4	Question: A spec of the DHT22 sensor is given in the link below. It mentions that the sampling rate is 0.5 Hz. https://lastminuteengineers.com/dht11-dht22-arduino-tutorial i) What does the sampling rate mean? ii) Where is this used in the Arduino code? Answer: i) Sampling rate means how often the sensor can produce a new measurement. A rate of 0.5 Hz means one new reading every 2 seconds ii) The interval is controlled by delay(15*1000); at the end of loop()
5	Question: Take a screenshot of your Serial Monitor displaying temperature & humidity sensor data logs. Add the image here.

	57.00,20.40 Answer: 57.00.20.40

Activity 2.2: Working with sensor - HC-SR04

HC-SR04 is an Ultrasonic sensor.

Hardware Required

Arduino Board

HC-SR04 Ultrasonic sensor

USB cable

Software Required

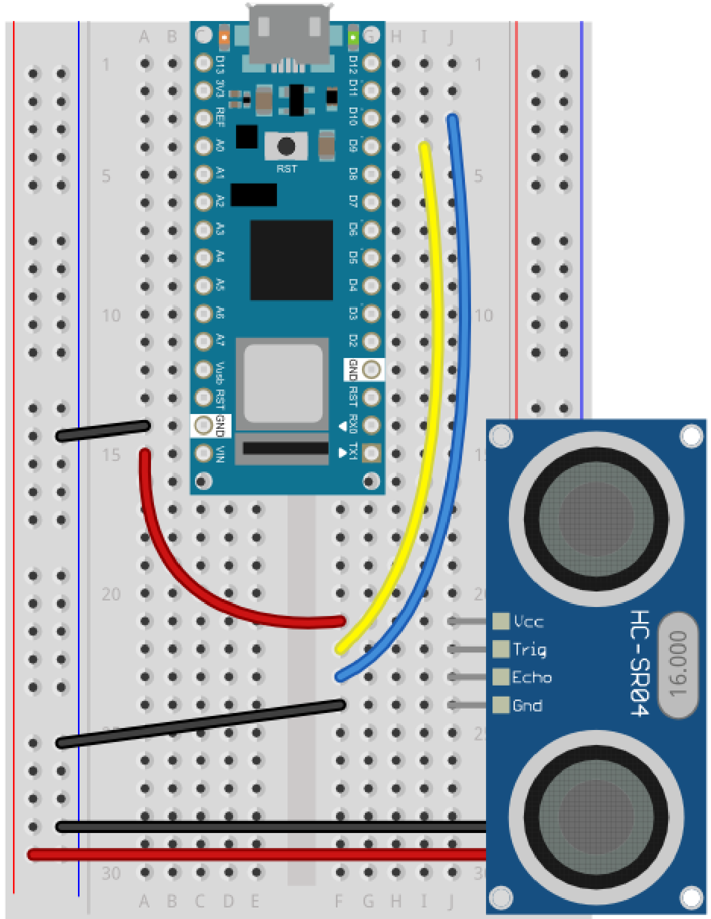
Arduino programming environment

Known issue, action required

The same known issue applies to SR04 which operates at 5 V source while the Arduino Nano 33 IoT supplies 3.3 V. It requires 2 VUSB (or VBUS) pads to be shorted to enable the pin.

Steps:

Step	Action
1	Connect your HC-SR04 sensor to the Arduino board. Note that the pin layout in the image below may look different than the board you may have.

	
2	Connect your Arduino board to your computer using the USB cable.
3	Write an Arduino sketch (or download it from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_2/sketch_hcsr04_distance.ino) which looks like below. Compile the code in Arduino IDE, deploy to the board and observe output in the Arduino IDE serial monitor.

```
const int trigger = 2;
const int echo = 3;

int getUltrasonicDistance(){
  // Function to retreive the distance reading of the ultrasonic
  long duration;
  int distance;

  // Assure the trigger pin is LOW:
  digitalWrite(trigger, LOW);
  // Brief pause:
  delayMicroseconds(5);

  // Trigger the sensor by setting the trigger to HIGH:
  digitalWrite(trigger, HIGH);
  // Wait a moment before turning off the trigger:
  delayMicroseconds(10);
  // Turn off the trigger:
  digitalWrite(trigger, LOW);

  // Read the echo pin:
  duration = pulseIn(echo, HIGH);
  // Calculate the distance in centimeter (CM):
  distance = duration * 0.034 / 2;

  // Uncomment this line to return value in IN instead of CM:
  //distance = distance * 0.3937008

  // Return the distance read from the sensor:
  return distance;
}

void setup() {
  // Define inputs and outputs:
  pinMode(trigger, OUTPUT);
  pinMode(echo, INPUT);

  // Start the serial monitor:
  Serial.begin(9600);
}

void loop() {
  // Print the distance to the serial monitor:
  Serial.print("Distance: ");
  Serial.println(getUltrasonicDistance());

  // Wait one second before continuing:
  delay(1000);
}
```

4

Question: Spec of SR04 is available here (https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf). Identify 2 critical aspects you should be careful about this sensor operation.

Answer: 1)The sensor needs a 5 V supply and the trigger pulse must stay HIGH for at least 10 microseconds.

2) Measure only within its stated range and keep the object inside the roughly 15 degree measuring cone

5	Question: Take a screenshot of your Serial Monitor displaying distance values while you try to generate a periodic motion by moving your hand gradually back and forth towards the sensor. Add the image here. Distance: 44 Distance: 7 Distance: 997 Distance: 2 Distance: 5 Distance: 7 Distance: 9 Distance: 13 Distance: 76 Distance: 995 Answer: Distance: 36

Activity 2.3: Working with sensor - Accelerometer

LSM6DS3 module on the Arduino Nano 33 IoT is an accelerometer and gyroscope sensor.

Hardware Required

Arduino Nano 33 IoT Board (has inbuilt LSM6DS3 module)

USB cable

Software Required

Arduino programming environment

Steps:

Step	Action
1	Write Arduino sketch (or download code from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_2/sketch_accelero.ino) which looks like below. Compile the code in Arduino IDE, deploy to the board and observe output in the Arduino IDE serial monitor.

```
// Add Arduino_LSM6DS3 library
// from Arduino IDE Library Manager.
#include <Arduino_LSM6DS3.h>

float x, y, z;

void setup() {
  Serial.begin(9600); // set baud rate
  while (!Serial);  // wait for port to init
  Serial.println("Started");

  if (!IMU.begin()) {
    Serial.println("Failed to initialize IMU!");
    while (1);
  }

  Serial.println(
    "Accelerometer sample rate = "
    + String(IMU.accelerationSampleRate()) + " Hz");
}

void loop() {
  // read accelero data
  if (IMU.accelerationAvailable()) {
    IMU.readAcceleration(x, y, z);
  }

  Serial.println(
    String(x) + ", " + String(y) + ", " + String(z));

  delay(1000);
}
```

2	Question: Spec of LSM6DS3 is available here (https://content.arduino.cc/assets/st_imu_lsm6ds3_datasheet.pdf). Identify at least 3 attributes of this sensor you think important to work with. Answer: 1) It can communicate using I2C or SPI 2) It combines a 3 axis accelerometer and a 3 axis gyroscope 3) The gyroscope has selectable ranges from +/-125 to +/-2000 degrees per second
3	Question: Take a screenshot of your Serial Monitor displaying sensor readings. Add the image here.

<table>
<tr><td></td><td>Answer: Accelerometer sample rate = 104.00 Hz
0.00, 0.00, 0.00
0.00, -0.00, 1.03
0.00, 0.00, 1.03
0.00, -0.00, 1.03
0.01, -0.00, 1.03
0.00, 0.00, 1.03
0.00, -0.00, 1.03
0.00, 0.00, 1.03
0.00, 0.00, 1.03
0.00, 0.00, 1.03
0.00, 0.00, 1.03</td></tr>
<tr><td>4</td><td>Question: Identify the max sampling rate and consider reducing the delay (line 37 in the sketch) to increase the number of samples. Summarise your findings here.

Answer: The sample rate at 104 Hz. The line delay(1000); prints only about one sample each second. Changing it to delay(10); gives about 100 prints per second, close to 104 Hz therefore a much smaller delay will not create new readings faster than the configured sensor rate</td></tr>
</table>

Activity 2.4: Plot data using Python Notebook

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. You can find detail in official website (https://matplotlib.org).

Hardware Required

No hardware required

Software Required

Python Jupyter notebook

Steps:

Step	Action
1	Download the Jupyter notebook **week2_notebook.ipynb** from here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_2/week2_notebook.ipynb). Follow the instructions in the notebook to carry out instructions and finally convert the notebook to PDF so you can combine it with this activity sheet PDF. Activity 2.4.pdf